

# Question Paper

Exam Date & Time: 20-Nov-2025 (02:00 PM - 05:00 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU  
FIRST SEMESTER M.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

**Advanced Systems Software [CSS\_5102]**

**Marks: 50**

**Duration: 180 mins.**

### PART-A

**Answer all the questions.**

- 1) Examine how different kernel synchronization techniques (spinlocks, semaphores, completions) manage concurrent access to shared data in kernel control paths. Justify which method is best suited for interrupt context and why. (5)
- A)
- Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Apply your understanding of process management to trace how a Linux process transitions between states such as Running, Sleeping, and Zombie. Describe a practical scenario (e.g., I/O wait or program termination) that triggers each transition (5)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Analyze how Linux manages page frames using the buddy system algorithm. Show with an example how page allocation and coalescing work to handle fragmentation. (5)
- A)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Evaluate how the Virtual File System (VFS) layer abstracts multiple file systems into a single unified interface. Explain how VFS data structures such as super\_block, inode, and file\_operations contribute to this abstraction (5)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) Analyze how splitting the kernel into modules improves maintainability and flexibility in Linux. Discuss how the kernel handles loading and unloading of device driver modules dynamically. (5)
- A)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) Demonstrate how Linux enables safe communication with hardware through I/O ports and memory-mapped I/O. Give one example of reading or writing to a device register using kernel I/O APIs. (5)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Demonstrate how modularization in the Linux kernel allows dynamic loading and unloading of device drivers. Describe the role of insmod and rmmod commands in this process. (5)
- A)
- Course Outcome: "CO4" / Cognitive Level: "Application"
- B) Apply your understanding and tell how the scull driver simulates a device using memory allocation techniques. Discuss how its read() and write() routines interact with kernel buffers. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Compare processes and threads in the context of parallel execution. Discuss how their memory usage and context-switch overhead affect performance in multi-core systems. (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Apply the concept of unification from the logic-programming model to demonstrate how a logic program matches variables to satisfy a query. Support your answer with a simple example showing how facts and rules are unified during execution. (5)

Course Outcome: "CO6" / Cognitive Level: "Analysis"

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| Event Name | Subject |
|------------|---------|
|------------|---------|

058-END SEMSTER-ODD-2025-PG CSS\_5102:Advanced Systems  
Software

| #     | Course Outcome | Description | Marks |
|-------|----------------|-------------|-------|
| 1     | 1              | CO 1        | 10    |
| 2     | 2              | CO 2        | 10    |
| 3     | 3              | CO 3        | 10    |
| 4     | 4              | CO 4        | 10    |
| 5     | 5              | CO 5        | 5     |
| 6     | 6              | CO 6        | 5     |
| Total |                |             | 50    |

\* **NA** - CO Not Selected

| Event Name | Subject |
|------------|---------|
|------------|---------|

058-END SEMSTER-ODD-2025-PG CSS\_5102:Advanced Systems  
Software

| #     | Level       | No of Questions | Marks (%) |
|-------|-------------|-----------------|-----------|
| 1     | Remember    | 0               | 0(0%)     |
| 2     | Understand  | 0               | 0(0%)     |
| 3     | Application | 3               | 15(30%)   |
| 4     | Analysis    | 7               | 35(70%)   |
| 5     | Create      | 0               | 0(0%)     |
| 6     | Evaluate    | 0               | 0(0%)     |
| Total |             | 10              | 50 (100%) |